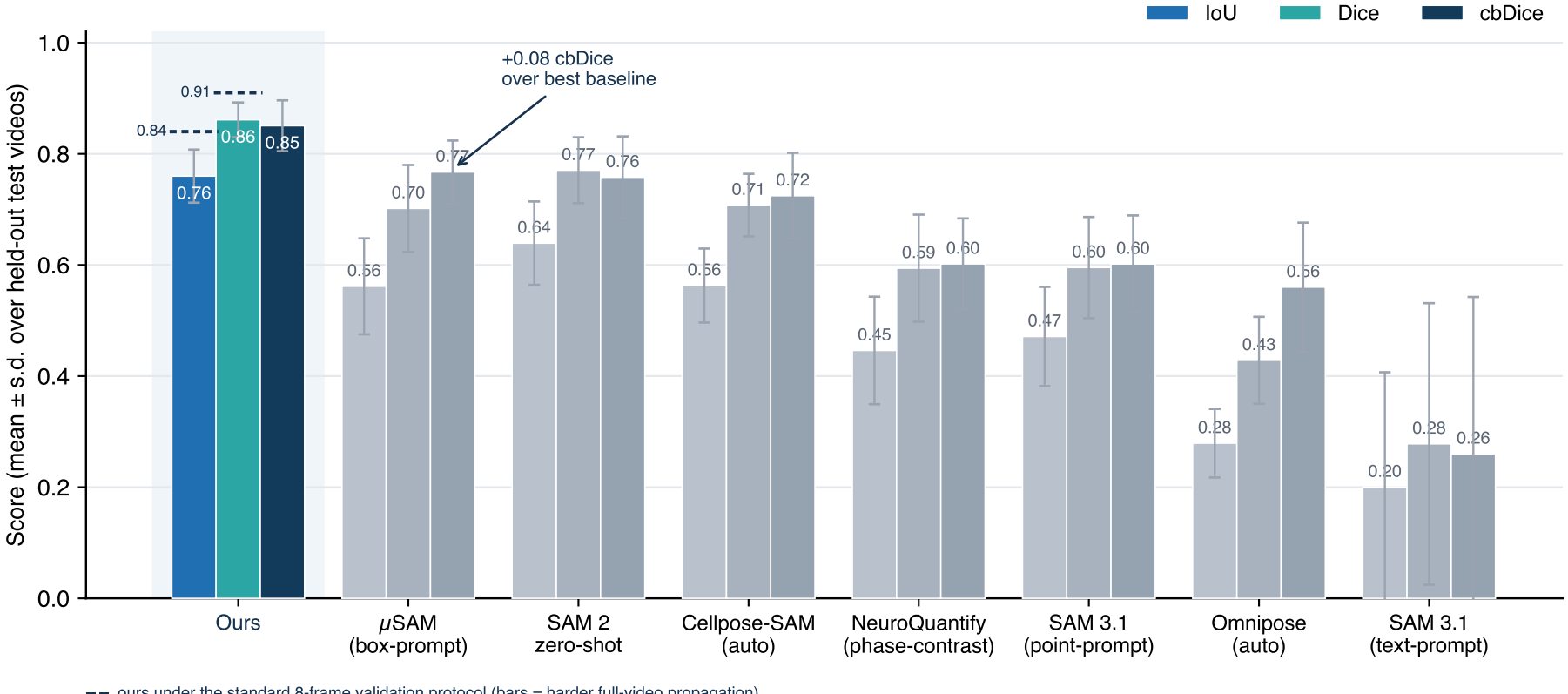
Neuron segmentation on the held-out test set — ours leads every metric, from a single frame-0 annotation
IoU
Dice
cbDice
Score (mean ± s.d. over held-out test videos)
1.0
0.8
0.6
0.4
0.2
0.0
+0.08 cbDice
over best baseline
0.91
0.84
0.76
0.86
0.85
0.56
0.70
0.77
0.64
0.77
0.76
0.56
0.71
0.72
0.45
0.59
0.60
0.47
0.60
0.60
0.28
0.43
0.56
0.20
0.28
0.26
Ours
μSAM
(box-prompt)
SAM 2
zero-shot
Cellpose-SAM
(auto)
NeuroQuantify
(phase-contrast)
SAM 3.1
(point-prompt)
Omnipose
(auto)
SAM 3.1
(text-prompt)
ours under the standard 8-frame validation protocol (bars = harder full-video propagation)